# Battery Degradation Testing - Internal Research Note

This note summarizes results from a comparative cycle-life study of two lithium-ion electrolyte formulations conducted by the materials team over a 4-month testing window.

## Test Methodology

Lithium-ion cells were cycled through 500 full charge-discharge cycles at a controlled ambient temperature of 25 degrees Celsius, using a constant-current constant-voltage charging profile at a 1C charge rate and 1C discharge rate.

Each formulation was tested across a batch of 24 cells to reduce the effect of manufacturing variance. Cells were rotated between test chambers every 100 cycles to control for chamber-specific effects.

## Capacity Retention

A cell was classified as end-of-life once its capacity fell below an 80 percent retention threshold relative to its rated capacity, consistent with the threshold used in the team's prior generation of cycle-life studies.

## Results

Cells using the new electrolyte formulation retained an average of 87 percent capacity after 500 cycles, compared to 74 percent for the baseline formulation under identical test conditions.

Internal resistance grew by 12 percent over the test window for the new formulation, versus 26 percent for the baseline, suggesting meaningfully slower degradation of the electrode-electrolyte interface over time.

## Thermal Behavior

No thermal runaway events were observed in either formulation across the full test matrix. Peak surface temperature during 1C discharge was 34 degrees Celsius for the new formulation and 38 degrees Celsius for the baseline.

## Next Steps

The team recommends proceeding to a larger-scale pilot of 200 cells using the new formulation, with an extended test target of 1000 cycles, before making a recommendation for production adoption.